

SHUN AKIYAMA

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Education

University of British Columbia

Expected Graduation: Nov 2028

Bachelor of Applied Science in Computer Engineering

Cumulative GPA: 4.33/4.33 (93.4%)

- Relevant Coursework: Software Construction, Computing Systems I & II, Algorithms & Data Structures, Circuit Analysis, Linear Algebra, Probability

Experience

UBC BizTech

Apr 2026 – Present

Software Engineer

Vancouver, BC

- Developing a full-stack hackathon and conference platform for UBC's largest technology club, serving **2000+** users and streamlining event registration, membership, payment, and real-time event workflows.
- Deployed a serverless backend with a **20-microservice** architecture using **AWS Lambda + Cognito** to manage registrations, payment, and event workflows
- Implemented Stripe webhook automation for **\$30,000+** in membership payments, reducing manual payment verification and triggering event confirmation workflows.

UBC Formula Electric

Sep 2025 – Present

Software / Firmware Engineer

Vancouver, BC

- Enabled in-vehicle firmware updates over CAN by implementing a **custom bootloader** with flash partitioning, CRC32 image verification, and linker-script memory layout configuration.
- Reduced multi-board firmware deployment time by **84%** by developing a parallel bootloading tool that coordinated flash erasure, firmware programming, asynchronous communication, automatic retries, and per-board progress tracking.
- Improved STM32 build consistency across **STM32F4/H5/H7** targets by contributing to a **Modern CMake** framework for configurable embedded builds, host tests, firmware packaging, and CubeMX validation.
- Strengthened fault detection across 6 STM32 ECUs by developing a watchdog library that refreshes the hardware watchdog only after **1 Hz, 100 Hz and 1 kHz FreeRTOS** tasks report healthy states.
- Accelerated board bring-up by developing Chimera, a Python HIL framework for remote GPIO, ADC, I2C, and SPI validation

Thompson Community Centre

Sep 2024 – Jul 2025

Badminton Instructor

Richmond, BC

- Coached 100+ beginner students ages 6–13, leading weekly classes of up to 13 and adapting instruction to different skill levels.
- Collaborated with staff to organize in-house tournaments that promoted teamwork, sportsmanship, and student engagement.

Projects

UBC Marketplace | TypeScript, React, Vite, Node.js, Express, PostgreSQL, Prisma, Supabase

July 2026

- Built a full-stack UBC student marketplace using React, Vite, Express, TypeScript, and Prisma, using a monorepo architecture with shared type-safe API contracts across the frontend and backend.
- Secured student-only marketplace access across 2 approved UBC email domains by integrating **Supabase Auth**, Express **JWT verification**, confirmed-email checks, and server-side authorization middleware.
- Designed a 7-model relational schema by modeling users, listings, images, favourites, reports, and listing-scoped conversations with Prisma, PostgreSQL, foreign keys, indexes, and uniqueness constraints.
- Built **real-time buyer-seller messaging** by synchronizing active chat threads, inbox previews, archive state, and unread counts with **Supabase Realtime** subscriptions and React state reconciliation.

Technical Skills

Languages: TypeScript, JavaScript, C, C++, Python, MATLAB

Web/Backend/Cloud: React, Vite, Node.js, Express, PostgreSQL, Prisma, Supabase, AWS Lambda, API Gateway, Cognito, DynamoDB, Stripe, WebSockets

Embedded/Firmware: STM32, ESP32, FreeRTOS, GPIO, I2C, SPI, ADC, CAN, bootloaders, linker scripts

Developer Tools: Git, CMake, STM32CubeMX, SEGGER J-Link, Postman